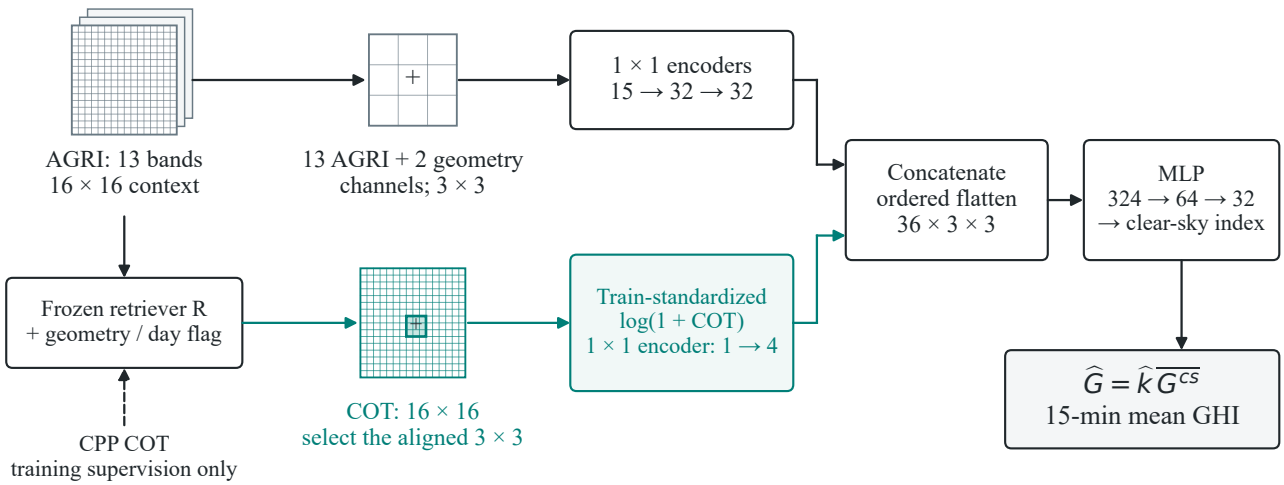
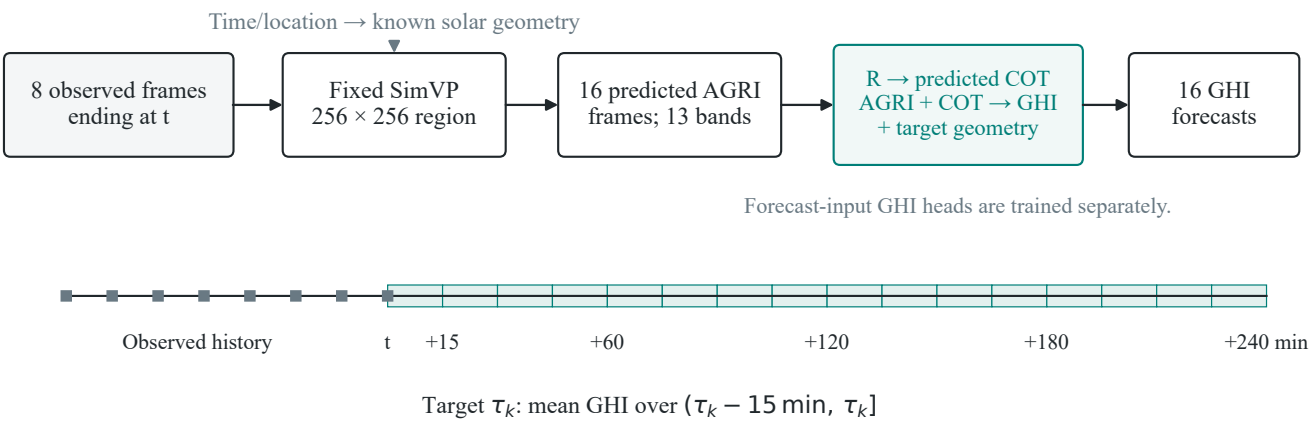

(a) COT-assisted estimation from observed imagery
AGRI: 13 bands
16 × 16 context
13 AGRI + 2 geometry
channels; 3 × 3
1 × 1 encoders
15 → 32 → 32
Concatenate
ordered flatten
36 × 3 × 3
MLP
324 → 64 → 32
→ clear-sky index
Frozen retriever R
+ geometry / day flag
CPP COT
training supervision only
COT: 16 × 16
select the aligned 3 × 3
Train-standardized
log(1 + COT)
1 × 1 encoder: 1 → 4
$\widehat{G} = \widehat{k}\,\overline{G^{cs}}$
15-min mean GHI
Nine spatial positions retained; no global spatial pooling.


(b) Future inputs from a fixed cloud-image forecast
Time/location → known solar geometry
8 observed frames
ending at t
Fixed SimVP
256 × 256 region
16 predicted AGRI
frames; 13 bands
R → predicted COT
AGRI + COT → GHI
+ target geometry
16 GHI
forecasts
Forecast-input GHI heads are trained separately.
Observed history
t
+15
+60
+120
+180
+240 min
Target τ_k: mean GHI over $(\tau_k - 15\,\text{min},\ \tau_k]$